PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION

FORM THREE TERMINAL EXAMINATION

CODE:031 **PHYSICS**

TIME: 3: 00 HRS **28th May, 2026**

INSTRUCTIONS

1. This paper consists of sections A, B and C with total eleven (11) questions.
2. Answer all questions in sections A and B and only two questions in section C.
3. All writing must be in blue ink except drawing which must be in pencil.
4. Cellular phones and any authorized materials are no allowed in the examination room.
5. Non-programmable calculators may be used
6. Write your name in all page
7. Where necessary the following constant may be used;
   (i) Acceleration due to gravity = 10m/*s*2 OR 10N/kg
   (ii) Density of water = 1g/*m*3 or1000kg/*m*3
   (iii) the coefficient of linear expansion of aluminum is **$2.6\times10^{-5}\ {}^{0}C^{-1}$.**

SECTION A (16 MARKS)
Answer all questions

1. For each of the following items (i-x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet provided.
   (i) Which of the following factors does NOT affect the pressure exerted by a gas in a closed container?
   A. Temperature of the gas
   B. Volume of the container
   C. Number of gas molecules
   D. Color of the gas
   E. None of the above

   (ii) According to Archimedes' principle, what determines the buoyant force acting on a submerged object in water?
   A. The weight of the object
   B. The volume of water displaced by the object
   C. The size of the object
   D. The density of the object
   E. Density of fluid

   (iii) Colours are produced when light passes through a glass prism because:
   A. Light waves interfere
   B. Glass prism colours the light
   C. In glass, different colours travel at different speed
   D. Different colours are filtered
   E. Diffraction of light occurs

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(iv) The laws of vector addition are;

A. The triangle law only
B. The triangle law and the polygon law only
C. The polygon law, triangle law and the parallelogram law
D. The parallelogram law.
E. Sine rule, cosine rule and Pythagoras theorem

(v) Which of the following combination of units represents a Joule

A.
B. Kgm/s2
C. Kgm2/s2
D. Kgm/s
E. Nm
F. Kg/m/s2

(vi) Action and reaction forces never cancel each other because

A. They act on different objects.
B. They are in the same direction.
C. They are parallel forces
D. They obey Newton's third law of motion
E. They act on the same object.

(vii) A material which allows some light to pass through it but one cannot see through it is said to be:

A. Transparent
B. Translucent
C. Luminous
D. Opaque
E. Colorless

(viii) Which of the following liquids does NOT contract if cooled from 4°C to 0°C?\

A. Milk
B. Pure oil
C. alcohol
D. Kerosene
E. water.

(ix) What physical phenomenon is observed when a tea bag is dipped into a cup containing hot water?

A. Surface tension
B. Capillary then diffusion
C. Diffusion
D. Osmosis then diffusion
E. Osmosis only.

(x) Which of the following is the correct weight of a body of mass 48 g when placed on the moon surface?

A. 0.48 N
B. 4.8 N
C. 0.80 N
D. 0.048 N
E. 80.0 N.

(xi) The temperature of a certain liquid is measured to be 300K. What will be its temperature in degrees centigrade?

A. 273°C
B. 100°C
C. 57°C
D. 37°C
E. 27°C

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2. Match the items in list A with the correspondence in list B by writing the letter of the correct response beside the item number.

LIST A	LIST B
(i) useful in drawing water from a borehole (ii) Applies the principle of first class lever (iii) It is velocity ratio increase by increasing the length of the ramp (iv)The effort is between load and fulcrum (v) Operates based on the principle of transmission pressure in liquids (vi)Carries objects from one point to another horizontally.	A Wheelbarrow B Wheel and axis C Hydraulic press D Tong E Inclined plane F. Claw hammer G. Pulley I. Windmill J. Screw jack

SECTION B (54 MARKS)

3. (a) Newton pair forces are equal in size and opposite in direction and yet they do not cancel each other out. Explain **(2marks)**

(b) A vehicle moves from rest with a uniform acceleration of 1.5m/$s2$ for the first 10s and continues accelerating at 0.5m/$s2$ for a further 20s. it continues at constant speed for 90s and finally takes 30s to decelerate uniformly to rest.

(i) Draw a graph of speed against time for the journey**. (3marks)**

(ii) From your graph, deduce the total distance travelled**. (4marks)**

4. (a) An Eskimo walking along an Iceland observed an inverted image in the sky of a polar bear standing some distance away. Explain **(2marks)**

(b) The refractive index for a ray of light travelling from air to water is 1.33. What is the refractive index for a ray travelling from water to air **. (3marks)**

(c) A light ray falls at normal incidence on the first face of an equilateral prism such that, the angle of incidence is 45^0, and emerges from the second face by making the same angle with the normal. What is the refractive index of the material of the prism? **(4marks)**

5. (a) (i) Define Inertia (**2 Marks)**

(ii) Explain why racing cars should have wide wheel tracks**. 2mark)**

(b) A uniform half metre rule is pivoted at its 30 cm mark. A mass of 50 g hung at the 45 cm mark keeps the rule horizontal. Determine the mass of the half-metre rule. (**5Marks).**

6. (a) Explain three characteristics of pressure in liquids**. (3marks)**

(b) A cube of side 2cm is completely submerged in liquid so that the bottom of the cube is at a depth of 10cm. If the density of the liquid is 1027kg/m^3 and g= 10N/kg.

(i) what is the difference between the pressure at the bottom of the cube and the pressure at its top? **(2marks)**

(ii) Determine the difference in the forces at the top and bottom**. (2marks)**

(iii)what is the weight of the water displaced by the cube? **(2marks)**

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7. (a) Give out three differences between vector quantities and scalar quantities**. (3marks)**

(b)**T**wo cars A and B are moving with velocities of 50 m/s and 30 m/s respectively. Find the relative velocity when they are in

(i) the same direction. (3marks)

(ii) the opposite direction**. (3marks)**

8. (a) (i)Explain in terms of electrons transfer, why a polythene rod becomes charged when rubbed With a duster? **(1mark)**

(ii) Explain why small pieces of paper are fist attracted toward a glass rod rubbed with silk but they immediately move away**? (2marks)**

(b) Two capacitors of capacitance2.5μF and 3.5μF connected in series. If the circuit is supplied by a potential difference of 20 v.

(i) Find the effective capacitance. **(2marks)**

(ii) The voltage on each capacitor **(2marks)**

(iii)The charge on each capacitor **(2marks)**

SECTION C (30 MARKS)

9. (a) (i) what is the anomalous expansion of water? **(2marks)**

(ii) Explain how fish and other aquatic organisms are able to survive in frozen lake**.(3marks)**

(b) (i) Show how coefficient of linear expansion and volume expansion for solid are related**. (5marks)**

(ii) The diameter of an aluminium sphere at 20°c is 2.5cm. If its diameter increases to 2.51 Cm when heated. What is the final temperature? **(5marks)**

10. (a) (i) what does the term coefficient of friction mean? **(2marks)**

(ii)Can the coefficient of friction be greater than one? Explain**. (3marks)**

(b) (i) Explain three methods of reducing friction**. (3marks)**

(ii)Starting from rest the time taken by a body to slide down a 45^{0} Inclined plane with Friction is twice the time it takes to slide down the same distance in the absence of Friction. Determine the coefficient of friction between the body and the Inclined plane**. (7marks)**

11. (a) Explain the following terms

(i) Electric current **(1mark)**

(ii) Current Electricity **(1mark)**

(b) Show that the two Resistor connected in series is given by $R_T = R_1 + R_2$ **(3marks)**

(c). (i) It is advisable to connect bulbs in parallel during electrical installation ,as an expert in physics explain the concept behind on this connection. **(4marks)**

(ii) why does a bird safely porch on a high potential electric wire? **(4marks)**

(iii)what does ohm (Ω) mean as applied in electric current. **(2marks)**

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